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A Framework for Evaluation and Use of Automated Scoring

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A framework for evaluation and use of automated scoring of constructed-response tasks is provided that entails both evaluation of automated scoring as well as guidelines for implementation and maintenance in the context of constantly evolving technologies. Consideration of validity issues and challenges associated with automated scoring are discussed within the framework. The fit between the scoring capability and the assessment purpose, the agreement between human and automated scores, the consideration of associations with independent measures, the generalizability of automated scores as implemented in operational practice across different tasks and test forms, and the impact and consequences for the population and subgroups are proffered as integral evidence supporting use of automated scoring. Specific evaluation guidelines are provided for using automated scoring to complement human scoring for tests used for high-stakes purposes. These guidelines are intended to be generalizable to new automated scoring systems and as existing systems change over time.

Keywords: automated scoring, essay scoring, performance testing, validity

Automated scoring of constructed-response items has transitioned from an area of academic inquiry into operational use for multiple high-stakes assessments and even more low-stakes learning environments. The time is ripe for reflecting on how such systems have come to be accepted and from this experience outlining a generalized framework for evaluating automated scoring systems for operational use in high-stakes environments. This paper proposes an overall framework to serve as a baseline approach to evaluate automated scoring for consequential assessment and to serve as a foundation for further expansion as our collective professional experience with such use grows. This paper provides a brief review of current systems for automated essay scoring and conceptual validity frameworks for automated scoring, as background for the introduction of the proposed framework, which will be illustrated through a discussion of how the framework is applied at Educational Testing Service (ETS) for the e-rater® automated essay scoring system. The framework also extends beyond direct evaluation of the automated scoring system itself and into the critical considerations for ongoing maintenance of automated scoring for high-stakes assessments.

Automated Scoring of Constructed-Response Tasks

More than 90 years ago the Army Alpha marked the first large-scale use of multiple-choice items in assessment, leading to the use of multiple choice items in nearly all large-scale high-stakes assessments. More recently the use of constructed-response tasks, in which an examinee produces a response rather than selecting from a range of provided options, has grown; within the last decade the TOEFL, SAT, and GRE have added constructed-response (speaking and/or writing) sections to the core test, and the ACT and TOEIC have added optional constructed-response sections. In addition, many licensure programs, including medicine, teacher licensing, public accounting, structural engineering, and architecture examinations, include tasks that require examinees to construct responses to tasks within the test. However, compared to their multiple-choice counterparts, such tasks take longer to administer with smaller contributions to reliability per unit time and delay score reporting because of the additional effort and expense typically required to recruit, train, and monitor human graders. The increasing use of constructed-response tasks stems from the belief that they measure aspects of a construct that are not adequately addressed through multiple-choice items. Although this argument may be more empirically supported for some constructs (e.g., Traub & Fisher, 1977, for verbal comprehension) and more debatable in others (e.g., Traub & Fisher, 1977, for mathematical reasoning; Lukhele, Thissen, & Wainer, 1994 for chemistry and history), even for constructs for which the unique contribution of constructed-response items to construct representation is disputable, the use of such items contributes an aspect of face validity which is desired by some stakeholders as being more consistent with authentic assessment (Linn, Baker, & Dunbar, 1991). Against this backdrop of increasing use the potential value of automated scoring, in which computer algorithms are used to score constructed-response tasks, is obvious. Such scoring, if designed appropriately, would enable a greater degree of construct representation and authentic assessment while providing performance that approximates some advantages of scoring multiple-choice items, including fast scoring, constant availability of scoring, lower scoring cost, reduced coordination efforts for human graders, greater

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score consistency, a higher degree of tractability of score logic for a given response, and the potential for a degree of performance-specific feedback that is not feasible under operational human scoring. This, in turn, may facilitate allowing some testing programs and learning environments to make greater use of constructed-response items where such items were previously too onerous to support. But of course, such potential advantages are not without challenges, including the cost and effort of developing such systems, the potential for vulnerability in scoring unusual or bad-faith responses inappropriately, and the need to validate the use of such systems and to critically review the construct that is represented in resultant scores.

Efforts to realize the potential advantages of automated scoring are far from new, with the development of automated scoring systems for essays originating with Project Essay Grade (Page, 1966). Project Essay Grade demonstrated how well even primitive automated representations of the construct of writing can predict the scores of expert human graders. Since that time, research on automated scoring systems has expanded. Fueled by the availability of ever more sophisticated delivery mechanisms, statistical methods, and technological innovations, there are currently many such systems designed to automatically score constructed-response tasks.

Some automated scoring systems may be classified as simulation-based assessments, which are characterized by task types that present computerized simulations of realistic scenarios for the examinee to complete, and for which the automated scoring systems are often not readily generalizable to other assessments, and sometimes even to other simulation-based tasks within the same assessment. Examples of such systems include the computer-based case simulations of standardized patients used in the United States Medical Licensing Examination™ developed by the National Board of Medical Examiners (Margolis & Clauser, 2006), the scoring of architectural designs constructed using a computer-aided design interface in the Architect Registration Examination (Williamson, Bejar, & Hone, 1999), the use of simulation-based assessment in the Uniform CPA Examination used in the licensure of Certified Public Accountants (DeVore, 2002), and in the scoring of simulations designed to measure the construct of information technology literacy in the iSkills™ assessment (Katz & Smith-Macklin, 2007).

Another class of automated scoring systems are response-type systems, which are designed to score a particular type of response that is in relatively widespread use across various assessments, purposes, and populations and therefore more readily generalizable than simulation-based scoring. Examples of such systems include essay scoring systems (Burstein et al., 1998b; Shermis & Burstein, 2003), automated scoring of mathematical equations (Singley & Bennett, 1998; Risse, 2007), scoring short written responses for correct answers to prompts (Callear, Jerrams-Smith, & Soh, 2001; Leacock & Chodorow, 2003; Mitchell, Russell, Broomhead, & Aldridge, 2002; Sargeant, Wood, & Anderson, 2004; Sukkarieh & Pulman, 2005), and the automated scoring of spoken responses (Bernstein, De Jong, Pisoni, & Townshend, 2000; Chevalier, 2007; Franco et al., 2000; Xi, Higgins, Zechner, & Williamson, 2008; Zechner & Bejar, 2006). Of these, the domain that has been at the forefront of applications of automated scoring has been the traditional essay response, which now offers more than 12 different automated essay evaluation systems for scoring and/or for performance feedback and improvement of writing quality. The most widely known of these systems include Intelligent Essay Assessor™ (IEA) by Knowledge Analysis Technologies™ (Landauer, Laham, & Foltz, 2003), e-rater® (Attali, & Burstein, 2006; Burstein, 2003), Project Essay Grade (Page, 1966, 1968, 2003), and IntelliMetric™ (Rudner, Garcia, & Welch, 2006). Each of these engines targets a generalizable approach to the automated scoring of essays, yet each takes a somewhat different approach to achieving the desired scoring, both through different statistical methods as well as through different formulations of what features of writing are measured and used in determining the score. Each of the most well known systems has commonalities as well as differences in their design and implementation. Each system has at the core of the capability a set of features that are designed to measure the elements of writing that are computer-identifiable and believed to be relevant to the construct of writing, even if they are not directly equivalent to what a human grader might identify in a similar effort. In this feature identification each system differs in that they use different sets of such features and, where features may overlap, likely use different and potentially proprietary approaches to the computation of those features. In addition, each system has in common the fact that it uses one or more statistical methodologies to derive a summary essay score from the feature values computed by the computer. However, these statistical methodologies differ by system, both in the choice of methodologies used and, when methodologies are held in common, by how they are applied within the capabilities.

Several of these systems are being used operationally to score high-stakes assessments in addition to being used as the engine for writing practice systems in educational settings. The first among these was e-rater (Burstein, Kukich, Wolff, Lu, & Chodorow, 1998a), which went operational for the GMAT® in 1999, with the GMAT program later transitioning to IntelliMetric (Rudner et al., 2006) as part of a shift of GMAT to a new vendor. The e-rater scoring system is also used operationally in conjunction with human scoring for the GRE Issue and Argument tasks (Bridgeman, Trapani, & Attali, 2009) since October of 2008, for the TOEFL Independent task since July of 2009 (Attali, 2009), and for the TOEFL Integrated task since November of 2010. Similarly, the IEA automated scoring engine is deployed in the Pearson Test of English used for high-stakes purposes. However, unlike e-rater, the IEA engine is used as the sole rater (Pearson, 2009).

The expanding use of automated scoring systems for high-stakes assessment underscores the fact that the fundamental question of automated scoring for such applications is no longer "Can it be done?" but "How should it be done?" That is, being a relatively nascent field of practice, what procedures are the current best practice for the evaluation and implementation of automated scoring? As the first to deploy automated scoring of essays for high-stakes assessment with the GMAT in 1999 and with multiple implementations in consequential assessments for other populations, such as TOEFL and GRE, ETS has developed a general framework for implementation in consequential assessment, outlined in the next section and expanded upon with illustrations of practice in subsequent sections.

Implementing Automated Scoring

Despite the operational implementation of automated scoring for high-stakes assessment across multiple programs, the field of study for automated scoring remains new enough that the guidelines of best practice are still evolving.

After a hiatus of almost three decades (Page, 1994; Page & Dieter, 1995; Page & Peterson, 1995; Shermis, Koch, Page, & Harrington, 1999), automated scoring work regained popularity in the early 1990s in response to the increasing demand for more efficient constructed-response scoring and the maturation of computer technologies. This has motivated the development of conceptual validity frameworks for automated scoring in the last decade (Bennett & Bejar, 1998; Clauser, Kane, & Swanson, 2002; Xi et al., 2008; Xi, 2008, 2010, in press; Yang, Buckendahl, Juszkiewicz, & Bhola, 2002).

An important trend in the conceptual work is the expansion of the scope of validity work for automated scoring to go beyond human-automated score agreement, considering the fact that automated scoring poses some distinctive validity challenges such as the potential to under- or misrepresent the construct of interest, vulnerability to cheating, impact on examinee behavior, and score users' interpretation and use of scores. These validity issues are either not relevant to human scoring or play out in very different ways.

The second important trend is a growing connection to Kane's argument-based approach to test validation (Kane, 1992, 2006), which helps establish a mechanism for combining different aspects of validity associated with automated scoring in an integrated fashion. Clauser et al. (2002) made the initial attempt to unfold validity issues regarding automated scoring in the larger context of a validity argument for the assessment. They presented some preliminary discussions of the validity threats that automated scoring would introduce to the overall interpretation and use of the resultant scores. Xi et al. applied a more elaborated framework for evaluating the use of automated scoring for a speaking practice test (Xi et al., 2008) and explicated the unique validity issues introduced by automated scoring to each of the inferences contributing to the overall validity argument for the assessment. Enright and Quinlan (2010) also evaluated the validity argument for using e-rater to complement human scoring for the Independent task in the TOEFL iBT Writing section drawing on Kane's argument-based test validation framework with an emphasis on the critical validity issues associated with implementing an automated system as the second rater.

Some of the work discussed above provides a conceptual analysis of areas of validity investigations associated with automated scoring (Clauser et al., 2002; Xi, 2010, in press; Yang et al., 2002) although others apply, refine, and expand a conceptual validation framework to a particular application of automated scoring (e.g., Enright & Quinlan, 2010; Xi et al., 2008). Nonetheless, none of them focuses on establishing the guidelines, criteria, and best practices under a unified evaluation framework for operational deployment of automated scoring in an assessment for high-stakes purposes. In this paper, we outline the key elements of the evaluation framework with some suggested criteria that may be useful to operationalize these elements as best practices in the evaluation and deployment of an automated scoring system as the second rater in assessments used for high-stakes decisions. Some of these methods and criteria may be easily generalizable to the evaluation of other applications of automated scoring (e.g., other automated essay or speech scoring systems that are combined with human scoring for high-stakes assessments) although others may be uniquely suited to the methods used in current applications of e-rater, which have been the focus of our efforts. However, even for instances in which the particular method may be unique to e-rater, the aspects of the framework these fulfill are appropriate for evaluation for other systems, though the specific methods and criteria may differ. We illustrate the fulfillment of the evaluation framework with the current applications of e-rater at ETS to help clarify the intent of the evaluation framework. We chose e-rater as the illustration for this framework because it is an automated scoring system extensively studied by multiple researchers and, being implemented in several consequential assessments, has been the focus of the development of guidelines and criteria for evaluating automated scoring. Since 1998, significant research and development efforts have been devoted to e-rater at ETS, as evidenced by more than 20 peer-reviewed publications that document various aspects of the quality of e-rater including construct representation, reliability, relationships with other criteria of writing ability, and consequences of using e-rater (see http://www.ets.org/research/capabilities/automated_scoring for a list of recent publications). A team of more than 15 research scientists, including natural language processing (NLP) scientists, computational linguists, cognitive scientists, and psychometricians, have contributed to e-rater development and evaluation, as have computer engineers, data analysts, and research assistants. Development is also overseen by technical and subject matter advisory teams consisting of psychometric and content experts. Given the capital and human resource investment in the development of such technology, and the complexity of the task it is designed to accomplish, it warrants some attention to ensure that such a system is performing as intended. Although illustrated with a single system, this framework is not based on a single instance of evaluation but represents the culmination (thus far) of research targeting multiple generations of e-rater as it has developed over time and been implemented in multiple high-stakes assessments. Given the extent of research conducted with e-rater and the current interest of the assessment community in automated scoring of essays, e-rater appears well suited as the example to illustrate the proposed framework. Despite our efforts in establishing and articulating the framework, the reader will recognize that several areas of best practice under the proposed framework remain relatively unexplored, and as such represent areas for further research.

The most straightforward representation of the framework consists of five areas of emphasis: construct relevance and representation; accuracy of scores in relation to a standard (human scores) (Davey, 2009); generalizability of the resultant scores; relationship with other independent variables of interest; and impact on decision-making and consequences of using automated scoring (see Table 1). These areas of emphasis correspond respectively to the explanation, evaluation, generalization, extrapolation, and utilization inferences in Kane's argument-based approach to test validation (Kane, 2006) and are consistent with the areas of investigations addressed in other recent conceptual and empirical work on automated scoring (Enright & Quinlan, 2010; Xi, 2008; Xi, 2010, in press). However, we have explicitly and systematically incorporated issues of fairness for subgroups of interest in the discussion of our framework, as fairness, defined as "comparable validity for *identifiable* and *relevant* groups" (Xi, 2010b, p. 154), is considered an aspect of the validity argument for an assessment (Xi, 2010). In particular, when applying the evaluation framework discussed above, we also examine subgroup

Table 1. A Summary of the Emphasis Areas, Guidelines, and Criteria

Emphasis Area	Corresponding Inference in an Argument-Based Validity Framework	Guidelines and Criteria
Construct Relevance and Representation: Evaluating the fit Between the Capability and the Assessment	Explanation	Construct Evaluation Task Design Scoring Rubric
Empirical Performance—Association with Human Scores	Evaluation	Human Scoring Process and Score Quality **Agreement with Human Scores** **Degradation from Human–Human Agreement** **Standardized Mean Score Difference** Threshold for Human Adjudication Human Intervention of Automated Scoring **Evaluation at the Task Type and Reported Score Level**
Empirical Performance—Association with Independent Measures	Extrapolation	**Within-Test Relationships** **External Relationships** **Relationship at the Task Type and Reported Score Level**
Empirical Performance—Generalizability of Scores	Generalization	**Generalizability of scores across tasks and test forms** **Prediction of human scores on an alternate form**
Score Use and Consequences—Impact on Decisions and Consequences	Utilization	**Impact on the Accuracy of Decisions** Claims and Disclosures Consequences of Using Automated Scoring

The guidelines in boldface are subject to analyses of subgroup differences for fairness.

differences, for example, in terms of relationships with human scores, generalizability of the resultant scores, and relationships with external variables of interest. Table 1 summarizes the emphasis areas of our proposed evaluation framework and the guidelines and criteria associated with each emphasis area, which will be discussed in detail in the following sections.

As argued by Xi (2010, in press), the method of automated scoring implementation and the intended use of the assessment scores (e.g., for high-, medium-, or low-stakes decisions) will determine the prioritization of the emphasis areas discussed above. Further, in determining whether there is sufficient evidence for each of the emphasis areas, different guidelines should be applied for evaluating the results given the intended use and the method of implementation.

There are a number of proposed models for the implementation of automated scoring. The implementation method chosen will depend on the consequences (and therefore risks) associated with the assessment program and the degree to which the automated scoring capability in question has succeeded in satisfying both construct and empirical criteria for use. In general, a versioned approach to capability development and application to operational programs should be encouraged. Under this versioning approach, the use of emerging automated capabilities will be restricted to low-stakes nonconsequential uses, although more mature and proven capabilities may be recommended for higher stakes applications if sufficient evidence of the appropriateness of the capability for that assessment is present to justify appropriate use. A rough ordering (from more conservative to more liberal use) of implementations for use of automated scoring is as follows:

- *Human scoring*. The standard baseline for comparison, typically the average of two human scores for consequential assessment, although for low-stakes programs a single human grader may be used. Pilot testing of automated scoring may be done as a "shadow" to this operational scoring.
- *Automated quality control of human scoring*. The results of a single human score and an automated score are compared. If there is a discrepancy beyond a certain threshold between the two then the response is sent to a second human grader. The reported score is based solely on the human score (either the single human score or the mean of the two human scores). This is the model implemented by the GRE program for the Issue and Argument tasks (Bridgeman et. al., 2009).
- *Automated and human scoring.* The score from a single human grader and automated score are averaged or summed to produce the reported score. Responses with score discrepancies beyond a certain threshold are scored by additional human graders. Proposed reporting policies vary, but adjudication procedures have included reporting the average of all scores provided, as well as reporting the average of the two scores in highest agreement, and several variations of these, conditional on the particular distribution of scores involved. This combination of automated and human scores with adjudication is implemented for the TOEFL Independent (Attali, 2009) and Integrated essays as well as for the GMAT Argument and Issue prompts (Rudner et al., 2006).
- *Composite scoring.* A variation on combining the summary values of human and automated scores is to treat the human evaluations as an additional feature (weighted appropriately) to combine with features derived from automated scoring to produce a composite score.
- *Automated scoring alone*. Reporting scores solely from the automated system. This is the most liberal use of automated scoring.

Obviously, the use of automated scoring for high-stakes decisions is subject to a higher burden of both the amount and quality of evidence to support the intended use than for

lower-stakes and practice applications. The choice of implementation policies for automated scoring would be influenced by the quantity and quality of evidence supporting the use of automated scoring, the particular task types, testing purpose, test-taker population to which it is applied, and the degree of receptivity of the population of score users to models of implementation. For example, different policies for the implementation of e-rater have been adopted by the TOEFL and GRE programs. This difference in policy was driven in part by differences in the test-taking populations and the nature of the test tasks, empirical prediction of impact of different deployment models on reporting scale, and market research expressing the confidence of score users in the quality of automated scoring.

Using current applications of e-rater at ETS as illustrative examples, the following sections demonstrate how this evaluation framework can be applied to real world applications of automated scoring. Concrete criteria for each of the aspects of the framework are proposed as examples of good practice to provide practical guidance for practitioners.

The e-rater automated scoring system uses a regression-based methodology to predict human scores on essays using a number of computer-analyzed features representing different aspects of writing quality. Once the regression weights are determined for these features, they can be applied to additional essays to produce a predicted score based on the calibrated feature weights. The current version of e-rater uses 10 such regression features, with 8 representing linguistic aspects of writing quality, and 2 representing content. Most of these primary scoring features are composed of a set of subfeatures computed from NLP techniques, and many of these have multiple layers of microfeatures that have cascaded up to produce the subfeature values. An illustration of the construct decomposition of e-rater resulting from this structure can be found in Quinlan, Higgins, and Wolff (2009). The scoring features of e-rater are mapped to the 6-trait model (Culham, 2003) commonly used to evaluate writing by teachers as described by Quinlan et al. (2009). This kind of construct map of the automated scoring technology is one way to address the construct relevance of automated scoring, the first area of emphasis in the proposed framework and the topic of the next section.

Construct Relevance and Representation: Evaluating the Fit Between the Capability and the Assessment

Automated scoring capabilities, in general, are designed with certain assumptions and limitations regarding the tasks they will score. Therefore, the initial step in any prospective use of automated scoring is the evaluation of fit between the goals and design of the assessment (or other use of automated scoring) and the design of the capability itself. As an illustrative example of this evaluation, consider that e-rater is designed to score essays primarily for linguistic quality of writing and that the representation of content within e-rater, although state-of-the-art for the field of automated scoring, is primitive compared to the abilities of human graders to understand content. This is illustrated in the construct map of e-rater provided in Quinlan et al. (2009), where the two content-oriented features are defined on the basis of typical patterns of word and phrasal usage. Therefore, one would expect that the construct embodied in an e-rater score would be more consistent with the construct of tasks that allow relatively unconstrained responses than for tasks that more highly constrain the permissible response. (An illustration of this effect is evident in the empirical results reported by Bridgeman et. al., 2009, for e-rater for the GRE Issue task, which is relatively unconstrained compared to the GRE Argument task. The Issue task requires test takers to present their own opinions on an issue whereas the Argument task requires an analysis and critique of a given argument on a topic with relevant reasons and examples). To evaluate the conceptual fit between the assessment and the automated capability the following steps are proposed:

- *Construct evaluation*. What is the match between the intended construct and the automated scoring capability? The construct of interest, as formally defined by the assessment program, is compared with that represented by the automated scoring capability. Similarities and differences, in both intent and operation, would be highlighted and summarized. A summary judgment of the conceptual fit between the scoring features of the automated system and the construct of interest would be provided. This evaluation would include the nature of claims that would be made about the construct as a result of the assessment, and the degree to which such claims are consistent with the construct embodied in automated scoring capabilities. Specifically, evaluations will be made about whether the scoring features are relevant to the construct of interest and cover key aspects of it.
- *Task design*. What is the fit between the test task and the features that can be addressed with automated scoring? This involves a review of the design of tasks and expected response types to ensure that they are consistent with automated capabilities. A summary judgment is provided for the conceptual fit between the test task and the automated capability on a task-by-task basis.
- *Scoring rubric*. Are the features extracted by the automated scoring mechanism consistent with the features in the scoring rubric? The scoring rubric used (or proposed) is reviewed to ensure that the characteristics called for in the rubric are consistent with the automated capabilities. A summary evaluation of the degree of fit is conducted, with emphasis on areas of mismatch and the implications of any such mismatch on the interpretation of scores.
- *Reporting goals*. Are the reporting goals consistent with the automated scoring capability? A review of the reporting goals of the program is expected to ensure that they are consistent with the automated capabilities being offered. For some capabilities and assessments it may be appropriate to report task-based performance feedback although for others task-level scores or even summed scores across tasks may be the most explicit scores that can be provided.

In these evaluations the stakes associated with the assessment are obviously a factor in the determination of whether automated scoring is a reasonable fit with the assessment. Higher stakes imply a more critical judgment of any discrepancies between the capabilities and intent of the assessment construct/task design. Similarly, practice tests or other low/no-stakes uses imply some leeway for more liberal interpretation of discrepancies presuming full disclosure to the users of such products. The intended implementation method also impacts the evaluations of the conceptual fit. Using automated scoring as the sole rater demands a much greater degree of congruence between the automated scoring features and the construct of interest than using automated

scoring in conjunction with human scoring. On the basis of such evaluations by knowledgeable researchers a summary set of recommendations and risks associated with use of automated scoring can be provided, setting the stage for the second criterion: empirical evaluations in comparison with human scores, discussed in the next section.

Empirical Performance—Association with Typical Scoring Method (Human Scores)

The emphasis of automated scoring evaluation has traditionally been on the agreement between the automated scores and human scores on the same response, which are typically considered the "gold standard," though we highlight problems with this perspective in the following section. This is a natural basis for evaluation because human raters are trained to apply the scoring rubrics reflecting targeted abilities, human scores are readily available, are based on the same response, and typically represent the best possible practical alternative to automated scoring. Furthermore, the use of human scores as the basis for optimizing statistical models of score production in most systems makes the relevance of human scores an explicit criterion of performance for automated scoring. The following are some of the acceptance criteria used for the evaluation of automated scoring with respect to human scores when automated scoring is intended to be used in conjunction with human scoring. At ETS these criteria are conjunctive, with any area of performance not meeting these criteria being flagged as a substantive concern. These criteria are based on evaluation of the human scoring process and score quality, agreement with human scores, potential degradation in agreement from human–human agreement, and mean score differences in the distribution of scores at the task level. In addition, thresholds for human adjudication and conditions for human intervention are defined. We understand that some assessments may include multiple tasks that use automated scoring so all pertinent evaluations should be made for scores aggregated across tasks as well.

- *Human scoring process and score quality.* Is the process used by human raters to score the performances and the quality of human scores adequate for training and evaluating automated scoring systems? This evaluation entails a review of the documented characteristics of the human scoring effort and an analysis of the quality of the human scores. The model building and evaluation process for automated scoring is largely dependent upon the quality of the human scores. Therefore, an evaluation of the adequacy of this process, from training through operational scoring (including QC checks such as regular rater calibration tests, rater performance monitor papers, and close monitoring by scoring leaders) is a relevant prerequisite for using these data in the development and evaluation of automated scoring models, especially because most automated scoring systems use these scores as the "gold standard" in calibrating the statistical models relating scoring features to the automated essay scores. In addition, the quality of the resultant human scores can be checked by looking at inter- and intrarater reliability.
- *Agreement of automated scores with human scores*. Does the agreement satisfy a predefined threshold? Agreement of automated scores with human scores has been a longstanding measure of the quality of automated scoring. Although it is common to report agreements as percentages of cases being exact agreements and exact-plus-adjacent agreements, we do not use this as a guideline of acceptance for automated scoring performance because of scale dependence (we would expect values to be higher by chance on a 4-point scale than on a 6-point scale) and sensitivity to base distributions (tendencies of human scores to use some score points much more frequently than others). Therefore, at ETS these are reported in statistical analysis reports as conveniences for laypersons rather than as part of acceptance criteria. Instead, we typically evaluate the agreement of automated scores with their human counterparts on the basis of quadratic-weighted kappa and Pearson correlations (Fleiss & Cohen, 1973). Specifically, the quadratic-weighted kappa[1] between automated and human scoring must be at least .70 (rounded normally) on data sets that show generally normal distributions, providing a threshold at which approximately half of the variance in human scores is accounted for by e-rater. This value was selected on the conceptual basis that it represents the "tipping point" at which signal outweighs noise in agreement. The identical criterion of .70 has been adopted for product–moment correlations with the same underlying rationale regarding proportion of variance accounted for by e-rater. It should be noted that the results from quadratic-weighted kappa and product–moment correlations are not identical, as kappa is computed on the basis of values of e-rater that are rounded normally to the nearest score point on the scale that the human graders use, whereas the correlation is computed on the basis of unrounded values (e-rater scores are provided unrounded so that when multiple tasks are combined for a reported score the precise values can be combined and rounded at the point of scaling rather than rounding before summation). It is worthwhile to note that because e-rater is calibrated to empirically optimize the prediction of human scores, the expected performance of e-rater against this criterion is bounded by the performance of human scoring.[2] That is, if the interrater agreement of independent human raters is low, especially below the .70 threshold, then automated scoring is disadvantaged in demonstrating this level of performance not because of any particular failing of automated scoring, but because of the inherent unreliability of the human scoring upon which it is both modeled and evaluated. Therefore, the interrater agreement among human raters is commonly evaluated as a precursor to automated scoring modeling and evaluation (and consistent with the construct evaluation as described in the previous sections of this paper).
- *Degradation from the human–human score agreement*. Is the difference between automated–human score agreement and human–human agreement below a predefined threshold? Recognition of the inherent relationship between the reliability of human scoring and the performance of automated scoring has resulted in another criterion of performance in relationship with human scores: degradation—specifically, that the automated–human scoring agreement cannot be more than .10 *lower*, in either weighted kappa or correlation, than the human–human agreement. This criterion prevents circumstances in which automated scoring may reach the .70 threshold but still be notably deficient in comparison with human scoring. It should be noted that in practice we have observed cases in which the automated–human agreement

for a particular task has been slightly less than the .70 performance threshold, but very close to a borderline performance for human scoring (e.g., an automated–human weighted kappa of .69 and a human–human kappa of .71), and we have approved such models for operational use on the basis of being highly similar to human scoring and consistent with the purpose of the assessment within which they are used. Similarly, it is relatively common to observe automated–human agreements that are *higher* than the human–human agreements for tasks that primarily target linguistic writing quality, such as GRE Issue and TOEFL Independent tasks (Attali, 2009; Bridgeman et. al., 2009).

- *Standardized mean score difference between human and automated scores.*[3] Is the standardized mean score difference below a predefined threshold? Another criterion for association of automated scores with human scores is that the standardized mean score difference (standardized on the distribution of human scores) between the human scores and the automated scores cannot exceed .15.[4] This criterion ensures that the distribution of scores from automated scoring is centered on a point close to what is observed with human scoring to avoid problems with differential scaling.
- *Threshold for human adjudication*. How much of a difference do we need to see between a human and automated score before another human rater is required? In implementing automated scoring, alternative thresholds for the definition of discrepancy when evaluating the agreement between automated and human scores may be considered. In human scoring it is common practice for most scoring scales in high-stakes programs that use double human scoring to consider scores that are one point apart (e.g., one judge issuing a 3 and the other a 4) to be in agreement under the interpretation that reasonable judges following the rubric may differ, especially when evaluating a borderline submission. Typically, when two human scores are considered discrepant an adjudication process occurs in which additional human graders are used and a resolution process followed to determine the final reported score. These adjudication and resolution processes vary substantially by program and are sometimes conditional on the particular distribution of initial human scores produced. In the implementation of automated scoring with precise values recorded (decimal values), a wider range of options are available for defining agreement, each of which has implications for the extent to which the results of automated scoring influence the final reported scores and therefore the ultimate evaluation of impact under the procedures defined above. The implementation policies of the GRE and TOEFL program represent an example of how programs may differ in their implementation of discrepancy policies as well as their use of the automated scores. The GRE program is using an "exact agreement" threshold for defining agreement between automated and human scores such that if the automated score is .5 or more different than the human score, the scores are considered to be discrepant and an additional human grader scores the submission. By contrast, the TOEFL program implemented a discrepancy threshold of 1.5 between the automated and human score as the minimum value that results in the scores being declared discrepant as this policy best represents the previous human score policy of a 1-point discrepancy being considered in agreement and a 2-point discrepancy to be discrepant, under normal rounding of the e-rater score. Both policies are the result of examination of policy impact on reported scores to ensure they are consistent with the scaled score distributions that clients are accustomed to as well as being consistent with the market research conducted on score user perceptions of automated scoring quality. Such policies of adjudicating discrepant scores are not only common among assessments with double human scoring, they are also common for other assessments, such as GMAT (Rudner et al., 2006) where one score is human and the other automated.
- *Human intervention of automated scoring.* What are the response characteristics that render automated scoring inappropriate? An automated scoring model relies on analyzing typical patterns of response characteristics to predict human scores and may not work well for responses that exhibit certain "abnormal" characteristics. Currently the e-rater technology will flag essays of excessive length or brevity, repetition, those with too many problems, or off-topic responses for scoring by human raters. This adds additional support for the quality of the scores produced.
- *Evaluation at the task type and reported score level*. What is the relationship between automated and human scores at the task type and reported score level? The evaluation of the impact of using automated scoring on the aggregate task type score (if applicable) and reported score for the writing section is even more critical than the evaluation at the task score level. At this step, we simulate the score that would result from substituting an automated score for a human score and determine the distribution of changes in reported scores that would result from such a policy. This poses an additional opportunity to compare the performance of scoring under the proposed model (automated and human) to that of the traditional model (two human graders). Each of these measures for evaluation in this section can be applied at multiple levels within an assessment. Specifically, these measures might be applied at the level of each individual task, at the level of a task type (e.g., GRE Issue task), and at the level of aggregate score (e.g., GRE Writing score, representing the aggregate of the Issue and Argument task scores). In typical practice at ETS, we first conduct the empirical associations with human score (agreement, degradation, and standardized mean score difference) at the task level. At the task type level (aggregated results across the individual tasks within the task type) and the reported section score level the entire contingent of measures discussed above are also employed in the evaluation of performance.

It should be noted that all the performance expectations in this section are based on the assumption that these evaluation criteria are being applied to a *different* set of data than the data used to build the automated scoring models. This requirement for an external data set is intended to represent a more generalizable measure of performance that would be more consistent with what would be observed on future data. Otherwise, if computed on the same data used for model calibration the measures of agreement for automated scoring would be inflated and, for a regression-based procedure like e-rater, the standardized mean score difference criterion would seldom be flagged. Further, it is assumed that if an automated scoring model is intended to be used on new tasks of the same task type, there should not be any overlap in tasks in the evaluation data with those in the model development data. Finally,

as pointed out by Xi (in press), in comparing the human-automated score agreement, the conditions under which the performance samples used for scoring model evaluation are produced should be consistent with the intended implementation method of automated scoring. Evaluation results based on mismatched conditions (e.g., using performance samples produced for human graders to evaluate the performance of an automated scoring engine that is intended to be used as the only rater in operational testing) may misrepresent the quality of the automated scores (Xi, in press). Test takers are very likely to adapt their test-taking behavior to the grader (e.g., focusing their attentional resources on the aspects that the grader can analyze) and may thus produce test responses that are much more challenging to grade by computers if they are aware of being scored by a computer only. This issue of mismatched conditions has largely been ignored in the previous literature on automated scoring and should be closely examined in evaluations of human–automated score agreement.

The criteria discussed above have been established for instances where an automated scoring system is used to either quality-control human scoring or as a contributing score in an assessment for high-stakes purposes. If automated scoring were to be used as the only rater to score a consequential assessment, more stringent criteria may be advisable.

In addition to associations with alternate ratings, the association with other independent measures represents another criterion for the empirical performance of automated scoring and is the subject of the next section.

Empirical Performance—Association with Independent Measures

The evaluation of automated scoring against the performance of human scoring is an interesting criterion. On the one hand it stands to reason that, because high-quality human scoring is the best viable alternative to automated scoring (or vice versa) and is the basis for building the statistical models for scoring within the automated systems, this would be a natural basis for evaluating the resultant quality of scoring. On the other hand, the problems and concerns with human scoring are well documented and represent a range of potential pitfalls including halo effects, fatigue, tendency to overlook details, and problems with consistency of scoring across time. Therefore, it is not unreasonable to expect automated scores, if they are to provide any advantages that overcome these concerns with human scoring, to at least occasionally produce scores that differ from human scoring. As a result, it is of relevance to investigate more than just the consistency with human scores and to also evaluate the patterns of relationship of automated scores, compared to their human counterparts, with external criteria (Attali, Bridgeman, & Trapani 2010; Petersen, 1997; Powers, Burstein, Chodorow, Fowles, & Kukich, 2002; Weigle, 2010). Such an expectation is certainly not new, as it is simply an extension of a classic question of validity, but with the focus not just on establishing a relationship between a score and some independent criterion, but on interpretation of potential differences in this relationship as indicative of relative merits or weaknesses of automated and human scoring.

These independent variables may be scores on other sections of the same test or external variables that measure similar, related, or different constructs. If human and automated scores reflect similar constructs, they are expected to relate to other measures of similar or distinct constructs in similar ways:

- *Within test relationships.* Are automated scores related to scores on other sections of the test in similar ways compared to human scores?
- *External relationships.* Are automated scores related to other external measures of interest in similar ways compared to human scores?
- *Relationship at the task type and reported score level.* Are the relationships similar at the task type and reported score level? These comparisons should be made both at the task/task type score level and reported score level.

In past practice for e-rater we have examined both within-test relationships through patterns of association with scores on other test sections, and external criteria, such as self-reported measures that may be of interest (e.g., self-assessments of writing ability, grades on essays in English class or on course papers, and grades on academic courses; Attali et al., 2010; Powers et al., 2002; Weigle, 2010). However, interpreting differences in patterns of association, when they do exist, may be somewhat problematic as the criteria available to us thus far are typically not direct external measures of exactly the same construct and are subject to the classic validity problem of being impure measures. For example, if the human scores on an essay have slightly lower correlations with the multiple-choice verbal section of a test than automated scores, is this an indicator that the automated scores are benefiting from the attention to detail and consistency of scoring to produce a more consistent measure or is it an indicator that the human scores are better capturing the aspects of the construct of ability that motivated the need for this aspect of the construct to be measured through constructed-response in the first place? In truth, however, we must admit that thus far in practice the fact that automated scoring models are calibrated to replicate human scores, and are typically quite successful at doing so, has resulted in there seldom being notable differences in association with external variables between automated and human scores that warrant such debate. However, for other domains of applications which are not as mature as automated essay evaluation, such as automated scoring of spontaneous speech, divergent patterns of correlations with external criteria have been found for automated scores versus human scores (Bridgeman, Powers, Stone, & Mollaun, 2012), which may be partially attributable to the limitations of current speech scoring technologies.

Although the association of scores with independent measures may inform questions of convergent and/or divergent validity measures, the question of reliability is addressed in part through generalizability measures, which is the framework component addressed in the next section.

Empirical Performance—Generalizability of Scores

Automated scoring has a great potential to improve the reliability of the reported scores. This is an empirical issue, though, and can be investigated in two ways.

- *Generalizability of scores across tasks and test forms.* How generalizable are the automated scores across tasks and test forms in comparison to human scores? How generalizable are the automated–human combined scores across test forms? A comparison of the generalizability of human

and automated scores across tasks and test forms will provide insights into how consistently students perform across tasks and test forms. For example, a G or Phi coefficient in the G theory framework can be computed using *task* as a random facet for human and e-rater scores, respectively, and the coefficients can be compared to indicate the extent to which the use of e-rater impacts the score reliability. Additional analyses should also be conducted based on automated–human combined scores, which are the reported scores for TOEFL or GRE Writing tasks. This would involve treating task and rating (human rating as the first rating and e-rater score as the second rating) as random facets and computing a G or Phi coefficient.

- *Prediction of human scores on an alternate form.* To what extent do automated, human, and automated–human combined scores on one test form predict human scores on an alternate form? This analysis will reveal whether the use of automated scoring may improve the alternate form reliability of the scores. Such analyses have been conducted for e-rater for scoring a TOEFL iBT Writing task, where e-rater was found to predict scores averaged across two human raters on alternate forms better than a single human score (Bridgeman, Trapani, & Williamson, 2011).

Beyond internal measures of consistency, a responsible implementation decision will also consider how use of automated scoring might impact the decisions and consequences associated with assessment scores.

Score Use and Consequences—Impact on Decisions and Consequences

Eventually scores from high-stakes tests are used to make important decisions about test takers. Tests used for high-stakes purposes are also expected to have significant consequences on their stakeholders. The use of automated scoring in high-stakes environments may have considerable impact on score-based decisions, and on test use, test preparation, instruction, and learning.

- *Impact of using automated scoring on the accuracy of decisions.* What impact does the use of automated scoring have on the accuracy of score-based decisions? In some contexts, assessment scores are used for classification purposes, for example, a binary decision about eligibility for admissions or exemption from English language coursework once admitted, or a decision regarding placing students into several levels of English class. Depending on the intended use of the assessment scores, the aggregated reported scores may be subject to further analyses to see if human–machine combined scores introduce a greater amount of decision errors than human scores.
- *Claims and disclosures*. What claims and disclosures should be communicated to score users to ensure appropriate use of scores? Researchers should work with the operational program to establish a common understanding of the intended claims and intent for disclosure of both strengths and limitations of automated scoring to ensure an informed population of score users. These claims and disclosures may include the extent to which different aspects of the target construct are covered by automated scoring and its major construct limitations. The strength of automated scoring, typically in terms of improving score reliability, should also be communicated to test takers and score users in an accessible way. Typically, some general statements about improved score reliability can be made available that target the nontechnical audience including test score users whereas concrete evaluation results are reported in research publications.
- *Consequences of using automated scoring.* What consequences will the use of automated scoring bring about? Replacing one human rater with automated scoring or using automated scoring to quality-control human scoring may change users' perceptions of the assessment, how users interpret and use the scores for decision-making, how test takers prepare for the test, and how the relevant knowledge and skills are taught. These are all important consequence issues that need to be further investigated after an automated scoring system is deployed. At ETS, research into the consequences of the use of e-rater to complement human scoring for TOEFL iBT Writing is well under way (Powers, 2011).

The interest in impact of implementation is not limited to overall groups, and because such impact might be differential by subgroup it is important to consider such subgroups of interest independently, which is the topic of the next section.

Fairness—Subgroup Differences

In considering fairness of automated scoring we have targeted the question of fairness to the direct question of whether it is fair to subgroups of interest to substitute a human grader with an automated score. This narrowing of the question of fairness makes the assumption that the task type is fair to all subgroups, to include in the assessment, and further, that the human scoring of responses is fair to all subgroups. Either of these assumptions may be debated on the basis of empirical evidence, but for the specific question of automated scoring, we have chosen to narrow the question to the specific area of inquiry comparing automated scores to their human counterparts. To address measures of fairness we propose investigating subgroup differences with regard to a few guidelines and criteria discussed above. The first is extending the flagging criterion of standardized mean score differences from the task level analysis discussed above to the question of subgroup differences (Ramineni, Williamson, & Weng, 2011). In so doing we have established a more stringent criterion of performance, setting the flagging criteria at .10, and applied this criterion to all subgroups of interest to identify patterns of systematic differences in the distribution of scores between human scoring and automated scoring for subgroups.

The second is examining differences in the associations between automated and human scores across subgroups at the task, task type, and reported score levels. Major differences by subgroups may indicate problems with the automated scoring model for these subgroups and should be evaluated for potentially undesirable performance with the subgroups in question.

The third is investigation of the generalizability of automated scores by subgroup. Substantial differences across subgroups may suggest that the scores are differentially reliable for different groups.

The fourth is examination of differences in the predictive ability of automated scoring by subgroup. This consists of two classes of prediction that are likewise related to the criteria and processes discussed above. First is to compare an ini-

tial human score and the automated score in their ability to predict the score of a second human rater by subgroup. The second type of prediction is comparing the automated and human score ability to predict an external variable of interest by subgroup. This approach is subject to the same concerns and ambiguities regarding identification of an appropriate external criterion previously discussed, and perhaps amplified as a result of the further emphasis on subgroup variations.

Finally, subgroup differences should also be investigated in relation to the decisions made based on the scores. This is the most prominent manifestation of group differences.

An example of a direct exploration of the question of fairness of automated scoring in the context of the GRE program can be found in Bridgeman et al. (2009) and a similar investigation for the TOEFL iBT test in Enright and Quinlan (2010).

Thus far, this paper has focused on the practical procedures and implications of approving automated scoring for operational use in a high-stakes testing program. However, this is only part of the consideration of going operational with automated scoring in that we must also consider the ongoing maintenance of such systems. Despite the fact that automated scoring of essays has reached a state-of-the-art that allows for high-stakes use, it is readily acknowledged that no system currently evaluates essays in exactly the same way that human graders do. As such, each of the current engines is subject to periodic changes as the state-of-the art of NLP continues to advance and new methods and capabilities for computer processing of text are developed and deployed. Therefore, along with initial implementation of automated scoring there needs to be a plan for the periodic reevaluation of performance as a result of such changes to the scoring engine. The following provides a brief outline of the evaluations undertaken at ETS to review and approve an engine upgrade to e-rater for operational deployment, which currently occurs on an annual basis. We provide this outline as an illustration of how the framework applies to engine upgrades in addition to implementation decisions for testing programs, as well as to emphasize that such automated scoring systems are more like software than fixed scoring technologies and therefore require periodic updating as the state-of-the-art of NLP and related technologies and procedures improve with further research.

Engine Upgrades

There are two steps in the evaluation and approval of a new scoring engine. The initial step is a conceptual review, in which the engine changes are evaluated to determine whether the proposed changes to the engine are conceptually meaningful and result in a better representation of the construct of interest. The subsequent step is an empirical review, in which the performance of the new engine is contrasted with the performance of the current engine, both directly and with respect to any differential associations with validity criteria of interest, which typically includes human scores as a convenient criterion but other external validity criteria are also relevant and desired in such an evaluation. Each of these evaluations, the construct and the empirical, are conducted both at the level of evaluating the individual NLP-based features that are candidates for inclusion in the engine in isolation as well as at the level of the full integrated engine. The specific steps of such an engine upgrade are as follows:

(1) *Component sign-off*. Each new or revised component, or module, of the engine requires documenting that the module meets the following criteria:
   (a) *Construct evaluation*. It should represent an improved conceptual representation of the construct in the design logic and computation procedures.
   (b) *Corpus review*. The new feature or component is evaluated against an existing corpus of cases to determine how it performs on annotated data. The characteristics of the corpus used should be described (e.g., origin, who performed any annotations, etc.) and contrasted with the target population that the module would be used within operational practice. This is the common source of "recall" and "precision" statistics commonly reported in NLP literature. Recall is defined as the proportion of cases identified by human annotators that are also identified by the automated system and precision is the proportion of cases identified by the automated system that were also identified by human annotators.

(2) *Engine evaluation*. A data set that is representative of the programs which use e-rater serves as a standardized data set for comparing the performance of the proposed new engine against the performance of the currently operational engine on the basis of the same empirical measures described above for evaluation of automated scoring for operational use. If the new engine performs as well or better than the former engine on these performance criteria *and* provides some strong argument for constituting a construct enhancement (i.e., better construct representation because of changes to or additions of engine components) then it is approved as the new version of e-rater.

(3) *Full model rebuild.* Once a new version of e-rater is approved, the new version will have added features and revisions to existing features that necessitate the rebuilding and reevaluation of all scoring models for operational writing tasks in the pool for each program. On the basis of the full set of reconstructed models, statistical reports are produced for each operational program to summarize the impact of the engine upgrade on the operational performance of the task pool, again with respect to the evaluation criteria addressed earlier in this paper.

Conclusion

This paper has provided a proposed generalizable framework for the evaluation of automated scoring for consequential assessment and illustrated key components of this framework through summaries of operational criteria, policies, and practices used at ETS in partial fulfillment of this framework for e-rater scoring of essays. It also outlines some important considerations for maintaining a program of use through future enhancements to automated scoring capabilities.

In this paper, we have oriented our discussion of the criteria, policies, and practices toward the use of automated scoring in combination with human scoring in high-stakes environments with brief discussions of how they might differ for other methods of implementing automated scoring (e.g., using automated scoring only). We have used e-rater as the example illustrating the framework in part because it is one of the most thoroughly researched automated scoring systems and also because of the current interest of the field in automated scoring of essays. However, the emphasis areas covered in our framework and the general principles adopted are cer-

tainly applicable to and useful for developing concrete guidelines for good practice for other applications of automated scoring.

There are many issues regarding use of automated scoring that are beyond the scope of this paper, but are no less important for operational testing programs. Issues requiring further investigation include the sustainability of the interpretations derived from the scores when scoring engines are refined or improved, the use of specific models of scoring when new tasks are added to the pool and their continued appropriateness, how large a sample of scored performances is required for adequate modeling of human scores, the use of automated scoring as examinee performances change over time for various reasons and implications for score equating, or the potential for human scorers to diverge from automated score predictions because of the changing composition of the human scorers such as better training, more or less experience, etc. Many of these important issues remain and warrant further investigation. Nevertheless, it is hoped that by presenting this conceptual framework for implementing automated scoring, and ETS procedures in satisfying this framework, this will serve as a useful benchmark for the professional community to continue the expansion and refinement of best practices toward a more complete set of empirical measures that more closely approximates the set of evaluation tools that are available to us for multiple-choice testing.

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Notes

[1]The equation for quadratic-weighted kappa is $K = 1 - \frac{E([X_1 - Y_1]^2)}{E([X_1 - Y_2]^2)}$, where it is comparing mean squared error between the pair of ratings that is supposed to agree (X_1, Y_1) and a pair of unrelated ratings (X_1, Y_2). When the variables have the same marginal distributions and an intraclass correlation of ρ, $\rho = \kappa$ (Fleiss & Cohen, 1973).

[2]Admittedly, a mismatch between the scoring rubric and the aspects of performance intended to be targeted by the assessment and inconsistency in human raters applying the scoring rubric introduces additional errors to the scores.

[3]We also examine the variance of the e-rater scores in comparison to that of the human scores, although we do not have a strict rule of thumb for what constitutes a large difference. Judgments about the acceptability of variance differences are made case by case by a technical review committee at ETS.

[4]Formula for standardized difference of the mean:

$$\overline{Z} = \frac{[\overline{X}_{AS} - \overline{X}_H]}{\sqrt{\frac{SD_{AS}^2 + SD_H^2}{2}}},$$

where $\overline{X}_{AS}$ is the mean of the automated score, $\overline{X}_H$ is the mean of the human score, SD_{AS}^2 is the variance of the automated score, and SD_H^2 is the variance of the human score.

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